

BANARAS HINDU UNIVERSITY

B.Voc. Semester IV Examination, 2023-24

Paper: BVCA-401 — Cloud Computing

Time: Three hours

Maximum Marks: 70

Instructions: Answer all sections. Candidates are required to write their answers in their own words as far as practicable. Symbols have their usual meanings.

SECTION — A

Question 1. Attempt all parts:**10 × 1 = 10 Marks**

- (a) Which of the following is a characteristic of PaaS?
- (a) Supports only one programming language
 - (b) Does not integrate with web services
 - (c) Provides an ability to “Auto-scale”
 - (d) Requires manual hardware updates
- (b) PaaS is primarily used for:
- (a) Networking hardware
 - (b) Developing, testing, running, and managing applications
 - (c) Providing email services
 - (d) Hosting web pages
- (c) Which model involves the infrastructure residing at the customer’s premises?
- (a) Public cloud
 - (b) Private cloud
 - (c) Hybrid cloud
 - (d) Community cloud
- (d) Which of these is an example of an IaaS provider?
- (a) Google App Engine
 - (b) Amazon Web Services (AWS)
 - (c) Salesforce
 - (d) Slack
- (e) SaaS applications are:
- (a) Hosted locally on user devices
 - (b) Accessible over the internet

- (c) Only for enterprise users
 - (d) Managed by end-users
- (f) Which service model allows users to avoid the cost and complexity of purchasing and managing physical servers?
- (a) SaaS
 - (b) IaaS
 - (c) PaaS
 - (d) NaaS
- (g) In which model do users have control over operating systems, storage, and deployed applications?
- (a) SaaS
 - (b) PaaS
 - (c) IaaS
 - (d) NaaS
- (h) SaaS applications often integrate with:
- (a) On-premise software only
 - (b) Other cloud-based services
 - (c) Local hardware solutions
 - (d) None of the above
- (i) Which deployment model offers resources over the internet to anyone willing to pay?
- (a) Public cloud
 - (b) Private cloud
 - (c) Hybrid cloud
 - (d) Community cloud
- (j) PaaS is advantageous for integration because it:
- (a) Requires less manual coding
 - (b) Provides pre-configured development environments
 - (c) Eliminates the need for middleware
 - (d) Reduces application complexity

SECTION — B

Question 2. Attempt any six parts (Answer in about 200 words each): $6 \times 6 = 36$ Marks

- (a) What are the advantages of using cloud computing?

- (b) What are the characteristics of cloud architecture that separate it from a traditional one?
- (c) What does Infrastructure-as-a-Service (IaaS) refer to?
- (d) What does the acronym SaaS mean? How does it relate to cloud computing?
- (e) What is the difference between block level and file level storage?
- (f) Explain the different types of cloud computing services.
- (g) What is the difference between different types of servers?
- (h) What are the basic requirements of a secure cloud software?
- (i) Discuss the use of a hypervisor in cloud computing.

SECTION — C

Long Answer questions. Attempt any two questions:

2 × 12 = 24 Marks

Question 3. What is Virtualization in Cloud Computing? Explain its types and benefits.

Question 4. What is RAID? How is it useful and explain its different types.

Question 5. What are the different types of Clouds in Cloud Computing? Explain with examples.

Question 6. Explain the role of cloud technologies for social networking. Discuss the operational and economic benefits of SaaS.